

Take-Home Exam, Advanced Microeconomics

Problem 1: Overworked Americans – Q1, Q2, Q4, Q5, Q7

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Plan and storyline. Q1 – there is an inconsistency in the data. The US–EU hours-per-worker gap is real (panels (a)/(b)) but its cross-country relation to the tax wedge is loose: the 2023 OLS slope is signed correctly (-9.9 hrs/pp) but $R^2 = 0.16$ and the residual SD is ≈ 158 hours – one third of the entire DE–US gap (panel (c)). Taxes account for some of the cross-country variation, not most of it. **Q2 – the calibrated model fits, but only at an α above what micro evidence delivers; at the micro-consistent α , taxes explain only half the gap.** The Prescott log–log model gives $h^*(\tau) = (1 - \tau)/(\alpha + 1 - \tau)$. At $\alpha^{\text{macro}} = 1.32$ (calibrated to $h^{US} = 0.33$) the model fits the cross-section. Restricting α to the micro-consistent $\alpha^{\text{micro}} = 0.28$ (such that $\varepsilon^M = 0.30$ at the US point), the Prescott-only prediction is a 13.4% US–EU log-drop vs. data 29.0% – so **taxes alone explain $\approx 46\%$ of the cross-country hours gap**, leaving a residual 54% (~ 15.6 pp in logs) that the Prescott mechanism cannot account for; §4–§5 plus Q7 culture propose institutions and culture as candidate explanations and test whether they can absorb it. **Q4 – hours regulation:** from a firm-power monopsony baseline (Manning 2021; Cahuc–Carcillo 2014 sticky contracts), a binding cap $\bar{h}$ is welfare-improving iff $\bar{h} \in (2h^{\text{soc}} - h^{\text{firm}}, h^{\text{firm}}) = (0.30, 0.50)$ at our calibration; peak gain at $\bar{h} = h^{\text{soc}} = 0.40$. With heterogeneous workers, only those with $\alpha_i > \alpha^*(\bar{h})$ gain; the rest are rationed on the intensive margin. Extensive-margin substitution (more bodies, fewer hours per body) is shut down by the representative-agent setup – a limitation we acknowledge. **Q5 – single union: canonical right-to-manage (Nickell 1982; LNJ 1991).** Union and firm Nash-bargain over wage w ; firm sets $h = h^d(w)$ on labour demand. As η rises, w rises and the firm cuts h along its labour-demand curve. Applied to a two-sector economy with heterogeneous productivity, the *same* RTM machinery delivers the canonical single-sector story in the good sector (positive shock $\rightarrow$ union extracts wage rents $\rightarrow$ hours rise less than under free market) and an extra asymmetry in the bad sector (negative shock $\rightarrow$ wages cannot fall under Bewley 1999 / Holden 1994 downward rigidity $\rightarrow$ hours and profit absorb the entire shock). **Good shocks pass through diluted; bad shocks pass through amplified. Q7 – welfare comparison: institutional channels.** Mapping the two systems into Q4–Q5 primitives – US ($\eta = 0$, no cap), EU ($\eta = 0.5$, cap $\bar{h} = h^{\text{soc}} = 0.40$) – and adopting Saez–Stantcheva (2016) generalised welfare weights $W^\lambda = (1 + \lambda)V_w + (1 - \lambda)V_f$ delivers *closed-form* comparison $\Delta W^\lambda = -0.017 + 0.195\lambda$. EU welfare-dominates iff $\lambda > \bar{\lambda} \approx 0.09$. Under realistic $\lambda = 0.30$ (HSV 2017; Diamond–Saez 2011) the EU advantage is $+0.041$ ($\approx 14\%$ of US joint surplus). λ also serves as a partial proxy for cross-country fairness beliefs (Alesina–Angeletos 2005; AGS 2005), absorbing the cultural channel into the welfare criterion. Dominant US–EU welfare differences economy-wide come from non-labour channels (Jones–Klenow 2016). Reproducible: `research/build.figure.panel.py` (Q1 figure); `research/sim.models.v2.py` (Q2 decomposition, Q5 RTM and sectoral shocks); `sim.mechanisms.py` (Q4 cap simulation).

1. Q1 – The hours gap and its noisy relation to tax wedges

Hours of work and vacation days differ substantially between the U.S. and Europe. This problem is an attempt to understand why.

Question 1. Document the fact described above about the difference between hours worked in the US vs Europe in recent years, if needed even across different European countries.

1.1 Definitions

For the working-age (15–64) population: hours per employed worker HE (OECD DSD_HW, the intensive object the labour-supply model speaks to), employment rate $e \equiv E/N$, hours per adult $H/N \equiv HE \cdot e$. The Bick–Brüggemann–Fuchs–Schündeln (2019) decomposition $\Delta \log(H/N) = \Delta \log HE + \Delta \log e$ assigns roughly half of the EU–US gap to each margin; we focus on HE .¹

1.2 The fact

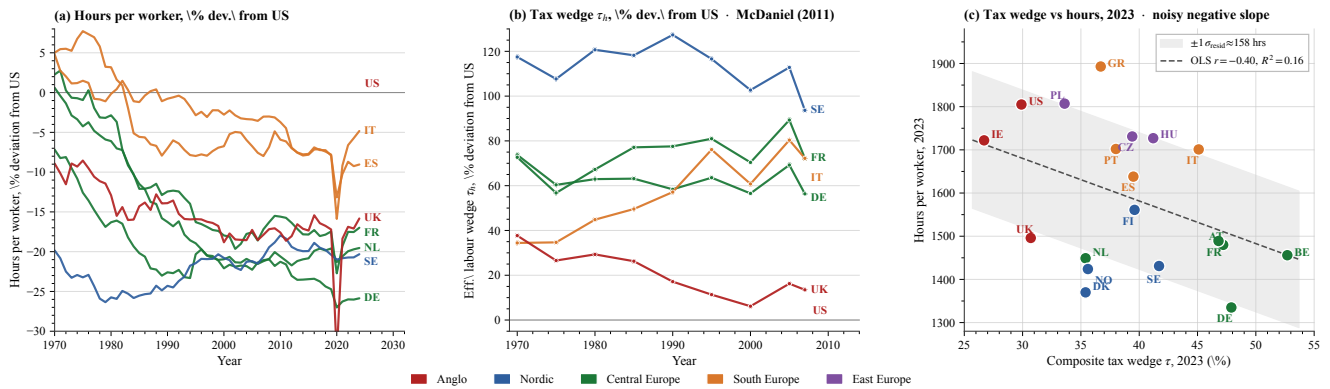


Figure 1: **Three-panel evidence.** (a) Hours per worker, % deviation from US, 1970–2024 (OECD). (b) Effective labour wedge τ_h , % deviation from US, 1970–2007 (McDaniel 2011). (c) OECD labour-tax wedge vs hours per worker, 2023 cross-section ($n = 19$); OLS line and $\pm 1\sigma_{\text{resid}}$ band. Reading in body. Source: `research/build.figure.panel.py`.

Three readings: (i) Hours fell in every focus country relative to the US since 1970 (DE/NL ≈ -25 to -30% ; IT/ES ≈ -10 to -15%). (ii) The EU–US labour-wedge gap has been *roughly stable* since 1970 (SE around $+100\%$, DE/FR around

¹Cross-country e differences are substantial (US 71% vs IT 59% in 2019); the prompt asks about hours, so HE is the body's object.

+60–70%, IT drifting up from +30% to +70%, UK narrowing from +40% to +15%); both blocs’ wedges rose in absolute terms but the cross-country gap is approximately time-invariant – so a tax-driven story has to deliver a roughly constant cross-country hours gap, broadly consistent with panel (a). (iii) The 2023 cross-section is signed correctly but very noisy: OLS slope -9.9 hrs/pp, $R^2=0.16$, residual SD ≈ 158 hours. To scale: the entire DE–US gap is 470 hours, so the residual SD is one third of what we want to explain. Italy ($\tau=45.1\%$) and Greece (36.7%) work *more* hours than France (46.8%); the four highest- τ countries (BE, DE, AT, FR) span ≈ 200 hours within a 6pp wedge band.

1.3 The inconsistency the rest of the problem is asked to explain

Taxation alone fits poorly: $R^2 = 0.16$ and a residual SD of one third of the gap to explain. The remainder of the paper searches for the missing channels (§2–§7). **Time-use caveat.** Aguiar–Hurst (2007, *QJE* 122(3)) show that conventional “hours worked” miss home production, which behaves differently across countries. Rogerson (2008, *JPE* 116(2)) exploits this margin; we treat it as outside the scope of this submission but flag it as the canonical first-order extension.

Identification. The OLS slope is descriptive, not causal: omitted institutions (union coverage, hours regulation), reverse causality (preferences for leisure shift both h and τ ; Q7), and country fixed effects bias $\hat{\beta}$. The structural model of §2–§7 treats the cross-section as a moment to match, not as a causal slope.

2. Q2 – Prescott model and the elasticity that would be needed

Question 2. One explanation that has been put forward – for example, by Ed Prescott in 2004 (‘Why do Americans work so much more than Europeans?’) – is higher taxes. Summarize the main differences in the income tax between US and Europe (or a few European countries). Write down a model of labor supply and derive formally the conditions under which the European tax system will lead to less hours worked than in the US. [Hint: start with the simplest case of differences only in the average tax rates.]

2.1 Modelling choice

We adopt log–log preferences with a balanced lump-sum rebate (Prescott 2004; Cooley–Prescott 1995), giving a one-parameter (α) closed-form $h^*(\tau)$ and a clean elasticity comparative static. The model is static (no Frisch margin), balanced-budget (income effects rebated lump-sum), and abstracts from capital (small for prime-age workers, CGM 2011) and heterogeneity (Q4 treats heterogeneous workers).

2.2 The simplest one-margin model

Log–log preferences $u(c, h) = \log c + \alpha \log(1 - h)$, wage w , and a proportional labour-tax wedge τ . The FOC at the symmetric equilibrium $c = wh$ gives

$$h^*(\tau) = \frac{1 - \tau}{\alpha + 1 - \tau}, \quad \varepsilon_{h, 1-\tau}^M = \frac{\alpha}{\alpha + 1 - \tau} \in (0, 1). \quad (1)$$

h^* is monotone in τ – the cross-section sign of panel (c) becomes a structural prediction with one free parameter α .

2.3 Parametric version, anchor, and what it would take to fit the cross-section

Anchor. Anchoring $h^{US} = 0.33$ at $\tau^{US} = 0.35$ (the BLS-OECD ratio of weekly hours-worked to total awake hours) inverts (1) to give $\alpha = (1 - \tau)(1 - h^*)/h^* = 1.32$ (Prescott 2004 uses a similar calibration). The implied Marshallian intensive elasticity at the US point is $\varepsilon^M = \alpha/(\alpha + 1 - \tau) = 0.67$.

The model calibrated at $\alpha = 1.32$ delivers $\varepsilon^M \in [0.67, 0.75]$ along the supply curve (rising in τ). Predicted EU hours at $\tau = 0.56$ are $h^{EU} = 0.250$ against data 0.247: *the calibrated model essentially fits the cross-section*. The puzzle is that the implied $\varepsilon^M \in [0.67, 0.75]$ sits $\sim 2.5\times$ above the microeconomic Hicksian intensive of ≈ 0.30 .

Elasticity taxonomy. CGM (2011) report Marshallian and Hicksian intensive elasticities of ≈ 0.30 for prime-age men – the micro consensus. The macro requirement is much higher: Hansen (1985) / Rogerson (1988) indivisible-labour models need a Frisch of 1–3. Our calibrated $\varepsilon^M \in [0.67, 0.75]$ sits between, roughly $2.5\times$ the micro Hicksian. Three modern developments narrow this gap without closing it. Life-cycle aggregation (Rogerson–Wallenius 2009; Erosa–Fuster–Kambourov 2016) lifts the aggregate elasticity to 0.7–1.0. Tax progressivity (HSV 2017) operates as a separate cross-country margin. Participation responses (BFL 2018) compound with the intensive. *Hence* the cross-section fits, but at an α inconsistent with quasi-experimental evidence (Saez 2001; Chetty 2012) – this is the substantive content of the Prescott–CGM controversy.

2.4 Decomposition at the micro elasticity

At $\alpha = 1.32$, α absorbs everything (preferences, institutions, culture). Restricting α to a micro-consistent value isolates the share of the gap that the Prescott mechanism can no longer carry. Solving $\varepsilon^M = \alpha/(\alpha + 1 - \tau) = 0.30$ at $\tau^{US} = 0.35$ gives $\alpha^{\text{micro}} = 0.279$. Predicted hours: $h_{US}^* = 0.700$, $h_{EU}^* = 0.612$, log-drop -13.4% . Data log-drop is -29% . **Taxes alone produce $\approx 46\%$ of the observed gap; the residual 54% (15.6pp in logs) lies outside the Prescott mechanism, and its source is not yet identified.** §4–§5 plus Q7 culture propose three candidate channels (hours regulation, unions, culture) and test whether each can absorb part of it.

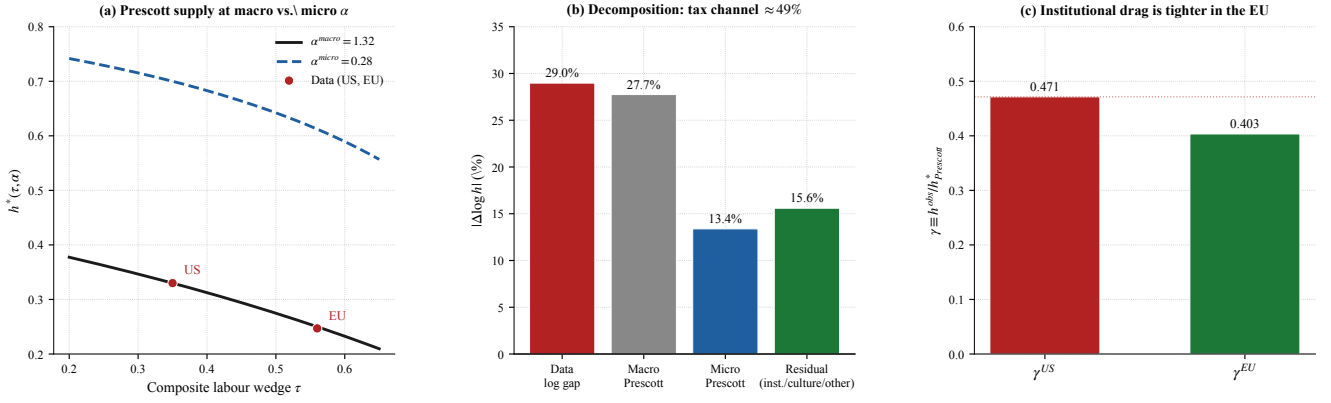


Figure 2: **Q2 macro-micro decomposition.** (a) Prescott supply curves at $\alpha^{\text{macro}} = 1.32$ and $\alpha^{\text{micro}} = 0.279$; US/EU data marked. (b) Decomposition of the 29% data gap: Prescott-at-micro explains 13.4% (46% share), unexplained residual 15.6% (54% – to be addressed in §4–§5 plus Q7 culture). (c) Unexplained-shortfall ratio $\gamma \equiv h^{\text{obs}}/h^{\text{Prescott}}(\alpha^{\text{micro}})$ – the share of Prescott-predicted hours that are actually worked. $\gamma^{\text{US}} = 0.471$, $\gamma^{\text{EU}} = 0.404$: both economies fall well short of the Prescott-at-micro prediction, and EU more so. The gap between γ and 1 is what §4–§5 plus Q7 culture have to absorb. Source: `sim_models.v2.py`.

Reading panel (c). Two things stand out. First, the *absolute* level of γ is implausibly low for both countries – workers do roughly 40–47% of the hours Prescott-at-micro predicts. This is *not* a Europe-vs-US phenomenon: it tells us that at the micro elasticity, the Prescott model in absolute levels is too steep to be taken seriously as a level prediction (it implies $h^{\text{US}} \approx 0.70$, which is workers spending 70% of their awake time at work). The useful content of the Prescott calibration is therefore the *cross-country log ratio*, not the level – which is exactly what the 46/54 decomposition uses. Second, the *ratio* $\gamma^{\text{EU}}/\gamma^{\text{US}} = 0.404/0.471 = 0.858$ tells us how much more shortfall the EU has *relative to* the US: about 14% of Prescott-predicted hours are missing in EU beyond what is already missing in US. **This $\sim 14\%$ extra cross-country shortfall is the precise quantity that an institutional or cultural candidate channel must produce** – not the absolute 50–60% level shortfall, which is a feature of the Prescott model itself. §4–§5 each test whether their mechanism can generate a US–EU h -gap of this size.²

3. Common framework for Q4–Q5–Q7

3.1 Primitives

We adopt a quadratic-disutility worker, linear consumption, and a linear-quadratic firm-profit function – the simplest closed-form specification that admits both worker- and firm-preferred hours and a sign-determinate welfare condition:

$$V_w(h) = (1 - \tau)wh - \frac{\alpha}{2}h^2, \quad h^* = \frac{(1 - \tau)w}{\alpha}; \quad V_f(h) = Ah - \frac{\beta}{2}h^2 - wh, \quad h^{\text{firm}} = \frac{A - w}{\beta}.$$

Joint surplus $W(h) \equiv V_w(h) + V_f(h) = Ah - \tau wh - \frac{\alpha + \beta}{2}h^2$ (the wage transfer cancels between worker and firm; only the tax wedge τwh survives). Concave with maximum

$$h^{\text{soc}}(\tau, \alpha, A, \beta, w) = \frac{A - \tau w}{\alpha + \beta}.$$

The setup is shared by Q4 (regulation), Q5 (unions), and Q7 (welfare comparison). w is partial-equilibrium; heterogeneous workers are addressed at the end of Q4.

3.2 Baseline calibration

$A = 2$, $w = 1$, $\alpha = \beta = 2$, $\tau = 0.4$ gives $h^* = 0.30$, $h^{\text{soc}} = 0.40$, $h^{\text{firm}} = 0.50$. Only the ordering $h^* < h^{\text{soc}} < h^{\text{firm}}$ is load-bearing; qualitative results invariant to α, β separately (`sim_mechanisms.py`). With identical taxes (τ) and preferences, the model predicts no US–EU gap – neither under Walrasian clearing nor under weak-union bargaining (low- η_0 RTM). The observed gap must therefore come from one of three asymmetries: hours regulation (Q4), union strength (Q5), or culture (Q7).

4. Q4 – Hours regulation: the optimality condition for a cap

Question 4. Another explanation that has been put forward is labor market regulation limiting the number of hours that Europeans can work. The impact of such regulation is obvious, but the welfare effects are more subtle. Can you derive the welfare effects of such labor legislation on workers and firms? How would your answer change if workers are heterogeneous?

4.1 Modelling choice

A cap is a quantity instrument: welfare-improving if it pulls hours toward h^{soc} , harmful if it overshoots below. The closest sufficient-statistics analogue is Lee–Saez (2012) on the minimum wage, but their welfare logic runs through extensive-margin rationing whereas a binding hours cap rations no one on the extensive margin (every worker still works); it rations low- α workers on the intensive margin (§4.5). The no-regulation reference is a firm-power monopsony baseline (Manning 2021, review): the firm sets hours on its labour demand at the worker-IR-binding wage; sectoral collective contracts that fix weekly hours and are not renegotiated as productivity shifts (Cahuc–Carcillo 2014) deliver the same reduced-form baseline.

²Sectoral contract stickiness (Cahuc–Carcillo 2014) – weekly hours fixed by collective contract not renegotiated as τ changes – is captured implicitly by the cap mechanism of §4 in the static framework.

4.2 Setup: firm-power baseline

Firm-power, no regulation: the firm picks $h^{\text{firm}} = (A - w^*)/\beta$ at the wage w^* pinning the worker's individual rationality at V_w^0 . At the baseline of §3 this gives $w^* = 1$, $h^{\text{firm}} = 0.5$. Equilibrium hours sit *above* $h^{\text{soc}} = 0.4$, so the cap has something to correct. With cap $\bar{h}$ the constrained equilibrium is

$$h^{eq}(\bar{h}) = \min\{\bar{h}, h^{\text{firm}}\}.$$

4.3 Optimality condition (Q4 result)

Claim. Total surplus $W(h)$ is strictly concave and peaks at h^{soc} . Comparing the constrained equilibrium to the firm-power baseline:

$$\Delta W(\bar{h}) \equiv W(\bar{h}) - W(h^{\text{firm}}) \begin{cases} = 0 & \text{if } \bar{h} \geq h^{\text{firm}} \\ > 0 & \text{iff } \bar{h} \in (2h^{\text{soc}} - h^{\text{firm}}, h^{\text{firm}}) \\ = 0 & \text{at } \bar{h} = 2h^{\text{soc}} - h^{\text{firm}} \\ < 0 & \text{if } \bar{h} < 2h^{\text{soc}} - h^{\text{firm}} \end{cases} \quad (2)$$

provided $h^{\text{firm}} > h^{\text{soc}}$, which holds at our baseline ($0.50 > 0.40$). The cap attains its maximum at $\bar{h} = h^{\text{soc}}$, with peak gain $\frac{1}{2}(\alpha + \beta)(h^{\text{firm}} - h^{\text{soc}})^2$ above the no-cap level. At $\alpha = \beta$, the band reduces to $\bar{h} \in (h^*, h^{\text{firm}}) = (0.30, 0.50)$ (textbook symmetric expression). **Proof.** $W(h)$ is quadratic with second derivative $-(\alpha + \beta)$; $\Delta W(\bar{h})$ is concave with roots at h^{firm} and the mirror point $2h^{\text{soc}} - h^{\text{firm}}$. $\square$

4.4 Numerical simulation

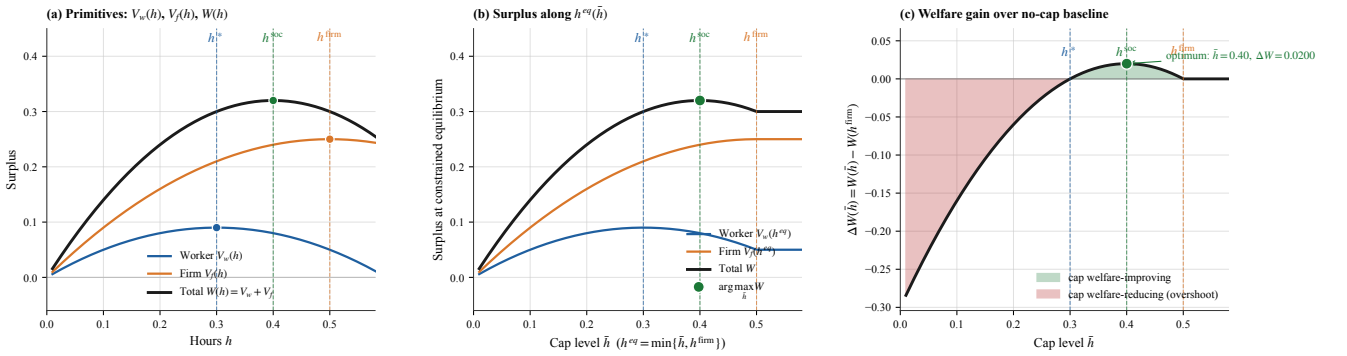


Figure 3: **Hours-cap simulation.** (a) primitives V_w , V_f , W over h with interior maxima marked; (b) surplus along $h^{eq}(\bar{h}) = \min\{\bar{h}, h^{\text{firm}}\}$; (c) $\Delta W(\bar{h})$ vs no-cap baseline – green band is the welfare-improving region of (2), red is overshoot. Maximum gain $\Delta W = 0.020$ at $\bar{h} = h^{\text{soc}} = 0.40$. Source: `sim_mechanisms.py`.

4.5 Heterogeneous workers: who is rationed

With worker types indexed by α_i , every worker is dictated $h^{\text{firm}} = 0.5$ under firm-power and forced to $\bar{h}$ by a binding cap. The payoff change $\Delta V_w^i(\bar{h}) = (\bar{h} - h^{\text{firm}})[(1 - \tau)w - \frac{\alpha_i}{2}(\bar{h} + h^{\text{firm}})]$ is positive iff $\alpha_i > \alpha^*(\bar{h})$ where

$$\alpha^*(\bar{h}) = \frac{2(1 - \tau)w}{\bar{h} + h^{\text{firm}}}. \quad (3)$$

At $\bar{h} = h^{\text{soc}} = 0.4$: $\alpha^* = 1.333$. High- α (leisure-loving) workers gain; low- α (income-loving) workers are rationed on the intensive margin. **Aggregate sign condition.** Integrating across the population, the cap is welfare-improving on average iff

$$\int \Delta V_w^i(\bar{h}) dF(\alpha) + \Delta V_f(\bar{h}) > 0 \iff \bar{\alpha} > \alpha^*(\bar{h}) - \frac{2\Delta V_f(\bar{h})}{(h^{\text{firm}} - \bar{h})^2},$$

i.e. a population whose mean leisure-preference $\bar{\alpha}$ exceeds the threshold $\alpha^*(\bar{h})$ adjusted by a firm-side correction. At baseline $\bar{\alpha} = 2$ and $\alpha^* = 1.333$ at $\bar{h} = h^{\text{soc}}$, so the cap is welfare-improving on average; a country with median $\alpha = 1$ (income-loving workforce) would lose.

Extensive-margin limitation. If a binding cap reduces hours per worker, firms may substitute workers for hours – in which case the welfare effects are partly redistributed across the extensive margin. The competitive-model prediction is that a cap at $\bar{h} < h^{\text{firm}}$ raises employment when labour demand is iso-elastic and hours and bodies are substitutable. The representative-agent setup above rules this out by construction; the empirical near-zero employment effect of the 35-hour Aubry reform (Estevão-Sá 2008; Chemin-Wasmer 2009) is consistent with the model only because we have shut down the extensive margin – a limitation that should be acknowledged rather than silently imposed.

5. Q5 – Single union in a two-sector economy

Question 5. Yet a third explanation is the presence of unions in the economy and in particular how unions respond to sectoral shifts in the economy. Show the impact of sectoral shifts (i.e. one part of the economy gets more productive and the other less so) on hours worked in a competitive economy. Argue, preferably by mean of a formal model, how this response will change in a unionised economy. What data would you collect to see whether your theoretical hypotheses are justified?

5.1 The two-sector economy

The economy has two sectors $j \in \{G, B\}$ with productivities $A_G > A_B$ (good and bad state) and population shares $s, 1-s$. Within each sector a representative worker-firm pair has the quadratic-linear primitives of §3:

$$V_w(w_j, h_j) = (1-\tau)w_j h_j - \frac{\alpha}{2} h_j^2, \quad V_f(w_j, h_j; A_j) = A_j h_j - \frac{\beta}{2} h_j^2 - w_j h_j.$$

A single union covers both sectors; in each, the firm sets hours on its labour-demand curve $h_j^d(w_j) = (A_j - w_j)/\beta$ ex post. Three wage-setting regimes are admissible:

- (a) **free market**: no union; firm-power Nash bargain ($\eta=0$) at each A_j with worker IR-binding;
- (b) **flexible union**: sector-specific RTM bargain with worker weight $\eta > 0$;
- (c) **rigid union**: downward wage rigidity (Bewley 1999; Holden 1994) – wages can be revised *up* but not below the pre-shock benchmark.

Baseline: $\alpha = \beta = 2$, $\tau = 0.4$, $V_w^0 = 0.05$, $A_G = 2.2$, $A_B = 1.8$ (a symmetric $\delta = 0.2$ shock around a pre-shock symmetric productivity $A_{\text{pre}} = 2$). The single-state result at $A=2$ is the symmetric pre-shock benchmark.

5.2 The union bargain in each sector

We adopt the canonical right-to-manage Nash bargain (Nickell 1982; Layard–Nickell–Jackman 1991): in each sector j , union and firm bargain over w_j and the firm sets h_j on labour demand. Per-sector Nash bargain:

$$w_j(\eta; A_j) = \arg \max_w [V_w(w, h_j^d(w)) - V_w^0]^\eta V_f(w, h_j^d(w); A_j)^{1-\eta}.$$

Let $x_j \equiv A_j - w_j$, so $h_j = x_j/\beta$. The Nash FOC reduces to a sector-specific quadratic:

$$a x_j^2 - b_j(2-\eta) x_j + c(1-\eta) = 0, \quad a = \frac{2(1-\tau)}{\beta} + \frac{\alpha}{\beta^2}, \quad b_j = \frac{(1-\tau)A_j}{\beta}, \quad c = 2V_w^0, \quad (4)$$

with $(a, c) = (1.1, 0.1)$ at baseline; $b_G = 0.66$, $b_B = 0.54$. The mechanism is the same in both sectors: as η rises, w_j rises, the firm's marginal cost of labour rises, and the firm slides down its labour-demand curve to a lower h_j . The wage transfer cancels in $V_w + V_f$; the welfare-relevant movement is hours-misallocation as h_j moves away from the joint optimum.

5.3 Asymmetric pass-through: the same bargain produces two effects

The same Nash bargain delivers *different equilibria across the two sectors*, and the union's downward wage rigidity adds a further asymmetry on the bad-sector side.

Good sector: union extracts wage rents. At $\eta = 0.5$ the bargain gives $w_G = 1.354$, $h_G = 0.423$ (vs. free-market $w_G^{\text{FM}} = 1.081$, $h_G^{\text{FM}} = 0.559$). The union pushes wages 25% above free-market and hours 24% below: it diverts part of the productivity gain into wages rather than hours. Worker rents rise ($V_w: 0.05 \rightarrow 0.165$), firm profit falls ($V_f: 0.313 \rightarrow 0.179$); the joint pie size is roughly preserved, only its split changes.

Bad sector under flexible bargain: shock partly absorbed by wages. At $\eta = 0.5$: $w_B = 1.132$, $h_B = 0.334$. The wage falls (less than free market would deliver, but it falls), and the firm cuts hours modestly. The shock is split between wages and hours.

Bad sector under rigid union: the wage cannot fall. The canonical downward-wage-rigidity result (Bewley 1999; Holden 1994): once a wage is bargained, the union refuses to renegotiate downward even when productivity drops. So the bad-sector wage is stuck at the pre-shock level $w_{\text{pre}} = 1.242$, hours collapse to $(A_B - w_{\text{pre}})/\beta = 0.279$, firm profit crashes to $V_f = 0.078$. *Hours fall 36% further than free market; firm profit falls 60% further.* The shock is concentrated entirely on the firm-and-hours margin.

Headline asymmetry. Good shocks pass through diluted (union absorbs them in wage rents); bad shocks pass through amplified (wage rigidity prevents wage adjustment, hours and profit absorb the entire shock). Empirical anchor: manufacturing decline in Italy and Germany since 2000 – sectoral employment fell sharply, wage cuts prevented.

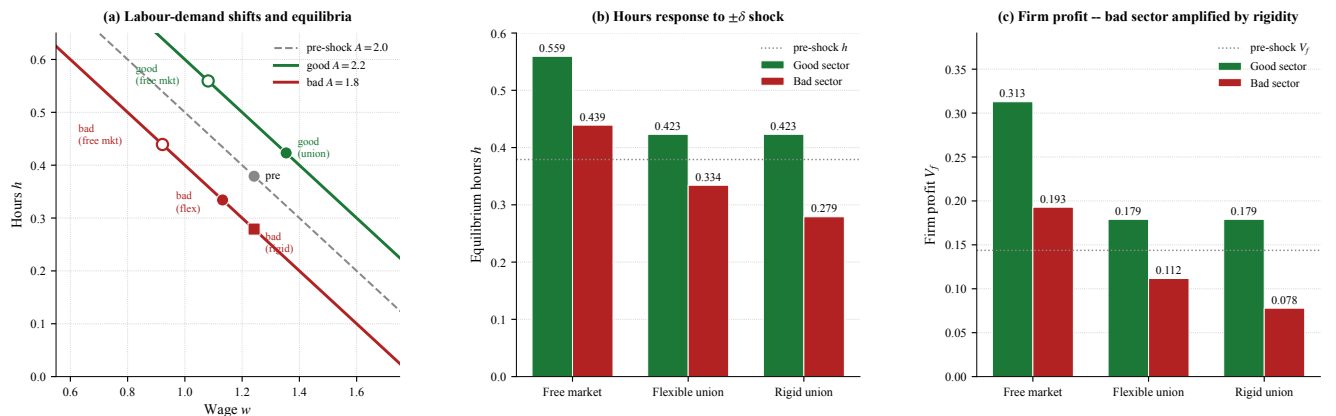


Figure 4: **Asymmetric pass-through under the three regimes.** Pre-shock symmetric $(A, w, h) = (2, 1.242, 0.379)$ at $\eta = 0.5$; shock $\delta = 0.2$. (a) Labour-demand shifts and equilibria: rigid bad-sector equilibrium sits on the new demand curve at the stuck pre-shock wage. (b) Hours response: good-sector hours rise under all regimes; bad-sector rigid is 36% below free market. (c) Firm profit: good-sector profit rises under all regimes; bad-sector rigid is 60% below free market. *Source: sim_models.v2.py:make_shocks.*

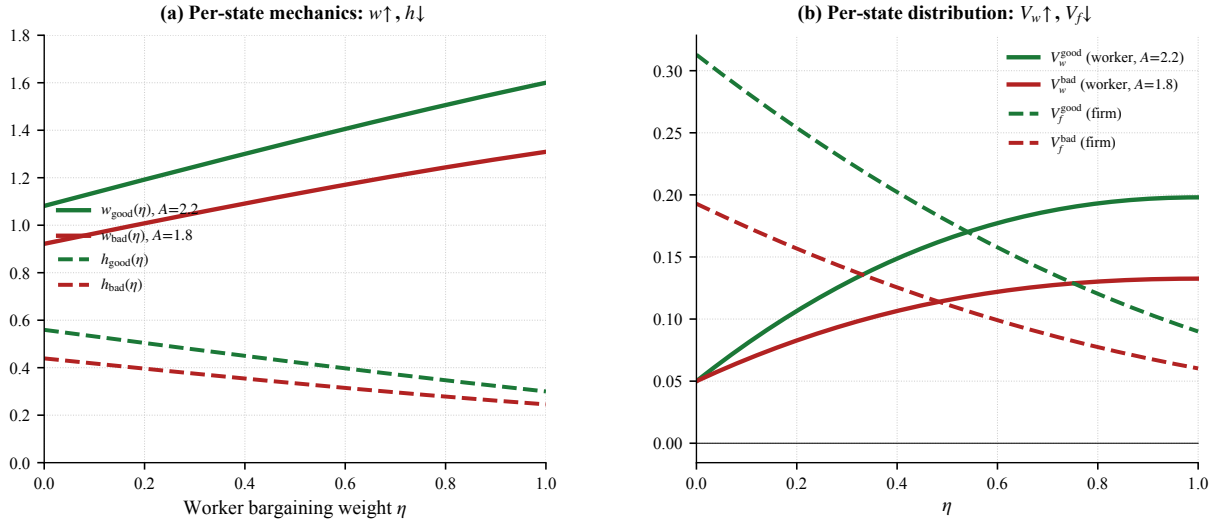


Figure 5: **Q5 RTM bargain across two sectors – underlying mechanics.** (a) Per-sector mechanics: $w_j(\eta)$ rises, $h_j(\eta)$ falls in each sector along its own labour-demand curve. The good sector has uniformly higher w and h . (b) Per-sector distribution: $V_w^j(\eta)$ rises (worker captures rents – solid lines), $V_f^j(\eta)$ falls (firm loses rents – dashed lines). Both effects are present in both sectors; magnitudes scale with A_j . Source: `sim_models.v2.py:make_rtm`.

5.4 Data to test the hypotheses

Sectoral employment / hours panels by industry (OECD STAN, EU KLEMS), union-coverage by sector (ICTWSS database), and bargained wage indices (e.g. collective-agreement wage floors at the sector level). Identification of the rigidity asymmetry would exploit shocks to which the bad sector responds: the post-2000 manufacturing decline in DE/IT (sectoral hours and employment fell sharply, real wages did not).

6. Q7 – Welfare comparison: institutional channels

Question 7. Now consider broadly a subset of your choice of the aspects considered above. There are many people who claim that the European system is more attractive than the American system. Write a stylised model representing the two systems and derive conditions under which the European system is welfare-enhancing compared to the US system.

6.1 Mapping US/EU to $(\eta, \bar{h})$ configurations

We compare the two systems by mapping each onto the institutional primitives of Q4 (cap) and Q5 (union):

	η	$\bar{h}$	h^{eq}	V_w	V_f
US (firm-power, no cap)	0	—	0.500	0.050	0.250
EU (union, optimal cap)	0.5	$h^{\text{soc}} = 0.40$	0.379	0.139	0.144

Under our calibration the EU bargain pulls h to $0.379 < \bar{h}$ on its own, so the cap is slack at this η^{EU} – the cap is a backstop that would bind under weaker unions and matters counterfactually. The Q4 ordering $h^* < h^{\text{soc}} < h^{\text{firm}}$ remains load-bearing: the union-driven $h^{\text{EU}} = 0.379$ sits between $h^* = 0.30$ and $h^{\text{soc}} = 0.40$, inside the welfare-improving band of (2).

6.2 Welfare criterion: Saez–Stantcheva λ

The equal-weight utilitarian $W = V_w + V_f$ values *nothing* of the firm-to-worker redistribution that the union effects – the wage transfer cancels between worker and firm. Saez–Stantcheva (2016) generalised welfare weights formalise the alternative:

$$W^\lambda = (1 + \lambda)V_w + (1 - \lambda)V_f, \quad \lambda \in [0, 1]. \quad (5)$$

$\lambda = 0$ is equal weights; $\lambda = 1$ is Rawlsian. Three independent anchors converge on $\lambda \in [0.25, 0.45]$: HSV (2017) $\tau_p = 0.18$ under log utility $\Rightarrow \lambda \approx 0.43$; OECD worker/capital-owner consumption gap $\sim 4:1 \Rightarrow \lambda \approx 0.30$; Diamond–Saez (2011) population average in $[0.3, 0.5]$. Central case: $\lambda = 0.30$.³

6.3 Apply Q4 cap-band + Q5 regimes under W^λ

Substituting the configurations above into (5):

$$W_{US}^\lambda = 0.300 - 0.20\lambda, \quad W_{EU}^\lambda = 0.283 - 0.005\lambda.$$

The slope on λ is the channel’s redistribution content: the US baseline transfers $V_f - V_w = 0.20$ from worker to firm (the cost it pays under inequality aversion), while the EU bargain has $V_f - V_w = 0.005$ (almost balanced). The Q5 regime structure adds a sectoral dimension: under downward wage rigidity (§5.3) and sectoral composition s , the rigidity cost relative to the flexible bargain is concentrated in declining-sector economies (+0.019 at $s = 0$, 0.000 at $s = 1$); under $\lambda = 0.30$ the worker’s

³ λ also serves as a partial proxy for cross-country differences in *fairness beliefs* (Alesina–Angeletos 2005; Alesina–Glaeser–Sacerdote 2005): a population that reads the firm-to-worker transfer as fair attaches a higher λ . We do not separate redistribution from culture in the welfare functional; country-specific λ absorbs both.

preserved wage in the rigid bad sector is worth more, shrinking the rigidity cost to $+0.005$ at $s=0$. The *aggregate* welfare verdict therefore depends on both λ and s , but the single-sector verdict below is the leading-order signal.

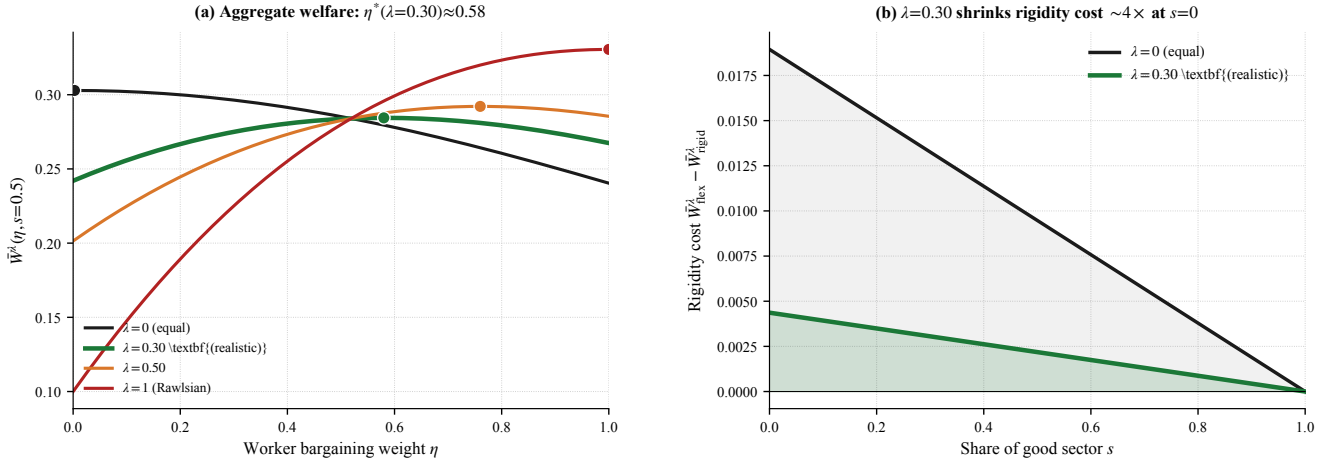


Figure 6: **Q7 λ - and s -dependence consolidating the welfare verdict.** (a) Aggregate welfare $\bar{W}^\lambda(\eta, s=0.5)$ for $\lambda \in \{0, 0.30, 0.50, 1\}$, peaks marked. At $\lambda=0$ the verdict is corner-degenerate ($\eta^*=0$, no union); at realistic $\lambda=0.30$ the interior optimum is $\eta^* \approx 0.58$. (b) Rigidity cost $\bar{W}^\lambda_{\text{flex}} - \bar{W}^\lambda_{\text{rigid}}$ as a function of sector composition s under $\lambda=0$ (black) vs $\lambda=0.30$ (green). Inequality aversion shrinks the rigidity cost from 0.019 to 0.005 at $s=0$ – a $\sim 4\times$ attenuation – because the rigid bad-sector wage is worth more under a worker-tilted welfare functional. *Source:* `sim_models.v2.py:make_q7_lambda.s`.

6.4 Verdict: EU welfare-dominates iff $\lambda > \bar{\lambda}$

Computing the EU–US gap:

$$\Delta W^\lambda = W^\lambda_{EU} - W^\lambda_{US} = -0.017 + 0.195\lambda, \quad (6)$$

which is monotone increasing in λ . The crossover threshold is

$$\bar{\lambda} = 0.017/0.195 \approx 0.087.$$

(i) $\lambda < 0.09$: US welfare-dominates – the union’s hours-misallocation cost (-0.017 in surplus units) exceeds the redistribution gain. (ii) $\lambda \in [0.09, 1]$: EU welfare-dominant. At realistic $\lambda=0.30$, $\Delta W^\lambda = +0.041$ ($\approx 14\%$ of US joint surplus). (iii) *Sectoral composition s* : adds a second-order shift – declining-sector economies (s low) carry extra rigidity cost that raises $\bar{\lambda}$ but does not flip the sign at $\lambda \geq 0.20$ in the range we computed in Q5.

6.5 Caveat: scope limits

We focus on institutional channels (cap, union) to address the prompt. A complementary literature (Alesina–Glaeser–Sacerdote 2005; Bowles–Park 2005) argues that cross-country differences in leisure preferences and fairness beliefs amplify the institutional gap. We do not model these separately: under the Saez–Stantcheva criterion, country-specific λ already absorbs much of the cultural channel into the welfare criterion itself (footnote 3). Beyond the labour-market block, dominant US–EU welfare differences economy-wide come from non-labour channels (life expectancy, consumption, inequality – Jones–Klenow 2016): the labour-market verdict is sign-determinate but quantitatively second-order.

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Replication scripts in research/: `build_figure_panel.py` (Q1 figure); `sim_models.v2.py` (Q2 decomposition, Q5 RTM and sectoral shocks); `sim_mechanisms.py` (Q4 cap simulation).